



Understanding Ground-Borne Vibration Propagation for Resilient Metro Infrastructure: Lessons from Delhi Metro's Expansion

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B O S T O N

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World-Class Metro Network

Passenger Safety

Reliable Infrastructure

Sustainable Mobility



Rail Grinding

Rail Profile Restoration

Improved Rail Performance

Extended Rail Life



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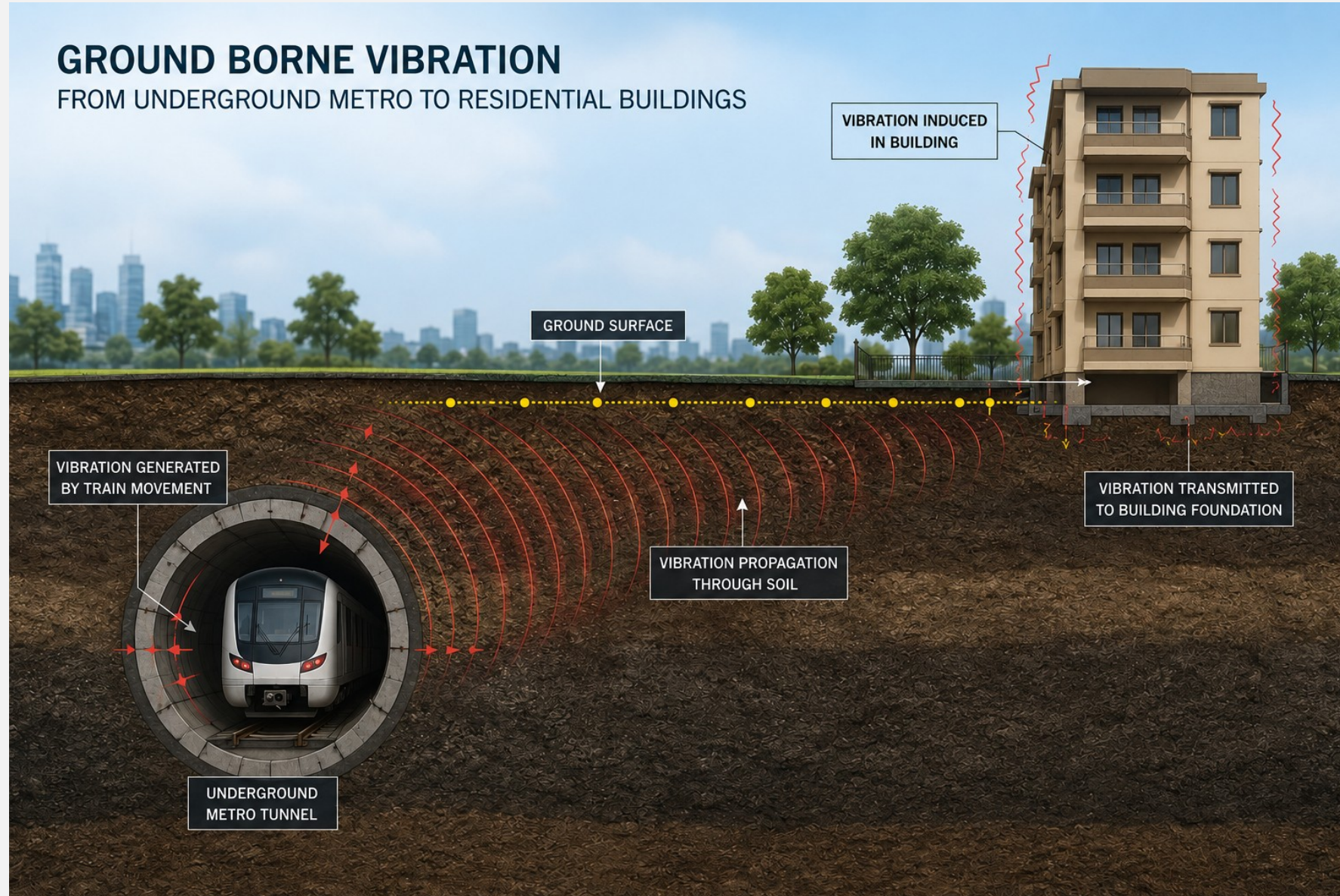
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Inferences and Post-Installation Measures

Introduction

GROUND BORNE VIBRATION

FROM UNDERGROUND METRO TO RESIDENTIAL BUILDINGS



- Source of ground-borne vibration → forces generated by wheel-rail interaction
- Vibrations are transmitted from the track structure through the tunnel lining into the surrounding soil.
- The vibration energy propagates through the ground and attenuates with increasing distance from the tunnel.
- A portion of the transmitted vibration reaches nearby building foundations and is further transferred into the building structure.
- The resulting vibration levels within buildings may influence occupant comfort.



Vibration Threshold Limits

Research Designs and Standards Organisation (RDSO) is the premier research and development body for Railways and Metros in India, which has circulated, Vibration Threshold limits for different Categories of Building under Guideline CT –38.

Building Category	Permissible Vibration Velocity (PPV)	Equivalent Vibration Decibel (VdB)*
Category –1	0.045 mm/s	65 VdB
Category –2	0.15 mm/s	72 VdB
Category –3	0.1 mm/s	75 VdB

Category 1 – High Sensitivity:

This Category includes buildings where vibration would interfere with operations within the building such as Hospitals, Research Institutes, Concert Halls.

Category 2 – Residential:

This Category includes all residential land uses and any buildings where people sleep.

Category 3 – Institutional:

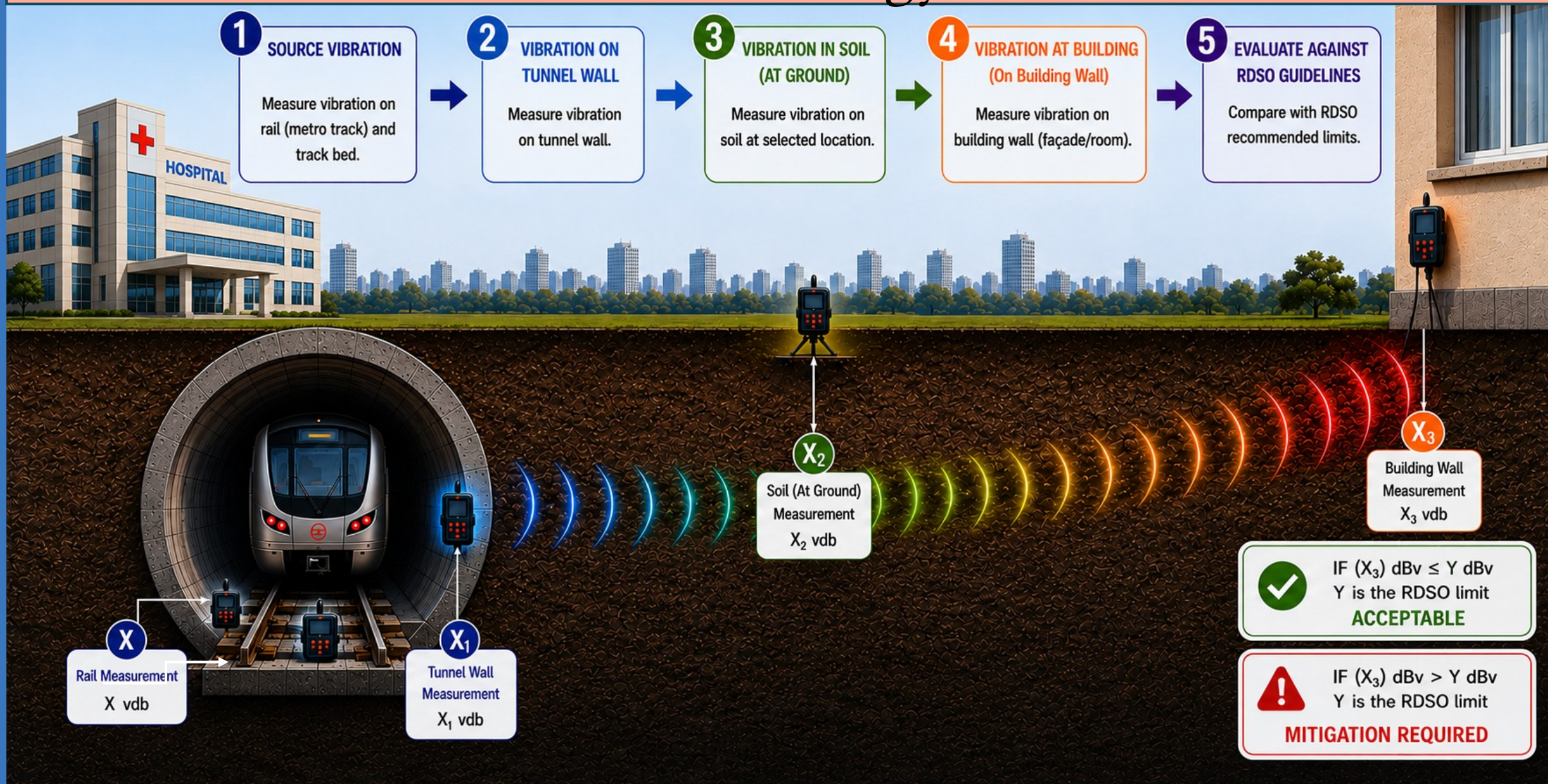
This category includes institutions, and quiet offices that do not have vibration-sensitive equipment

$$\text{Vibration Decibel (VdB)} = 20 \log_{10} \frac{V}{V_{ref}}$$

where, $V_{ref} = 1 \text{ microinch/sec or } 2.54 * 10^{-5} \text{ mm/sec}$



Methodology





Under Ground Corridors

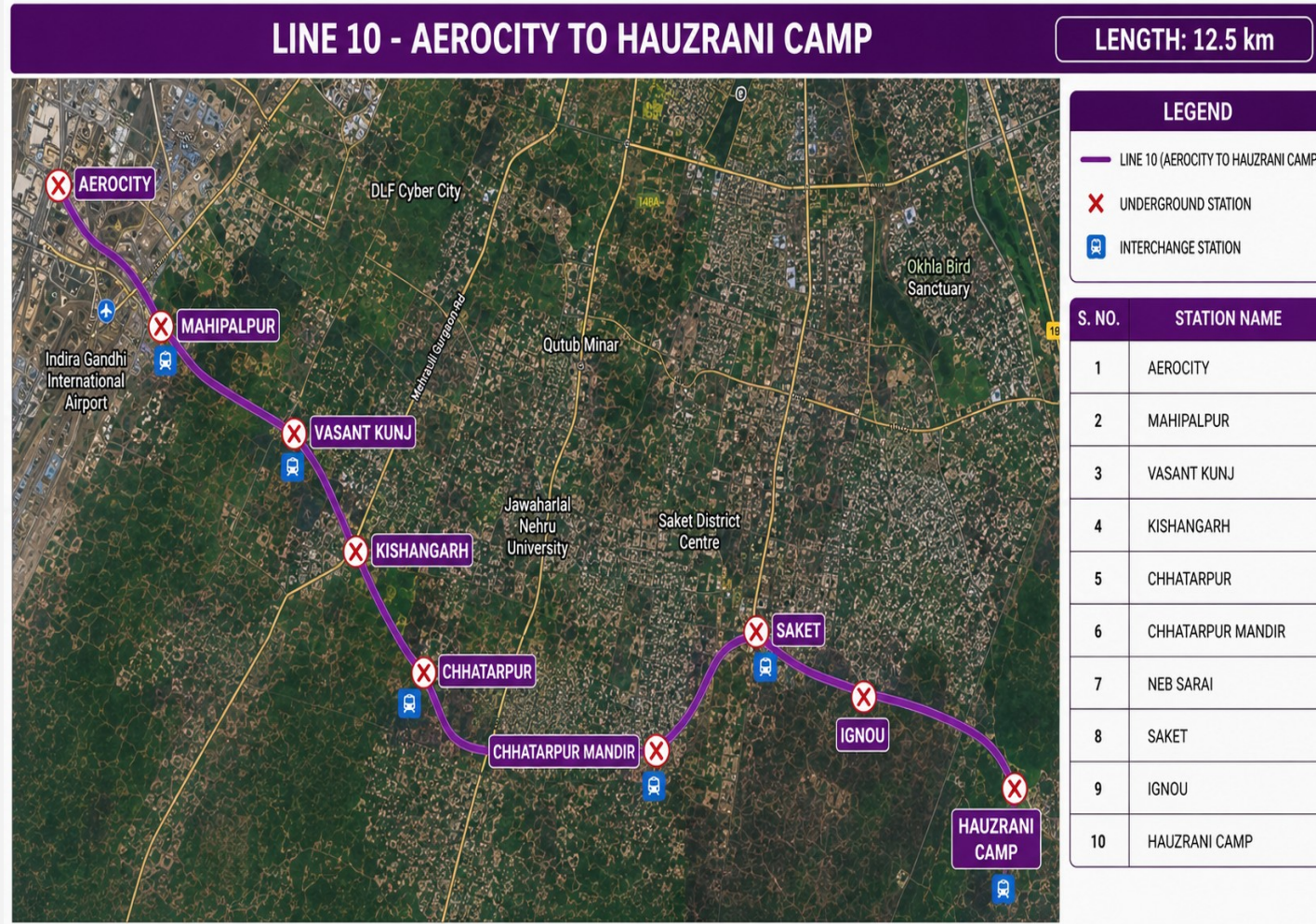
Line -08 (Derawal Nagar to R. K. Ashram) : Key Engineering Considerations



- Total Length: 7.4 Route Kms
- Alignment passes through highly congested and densely built urban areas
- Presence of old, fragile structures
- Critical stretch: Nabi Karim to Sadar Bazar with sensitive buildings
- Required **enhanced vibration assessment and targeted mitigation planning**

Under Ground Corridors

Line-10 (Aerocity to Hauz Rani Ramp) – Key Engineering Considerations

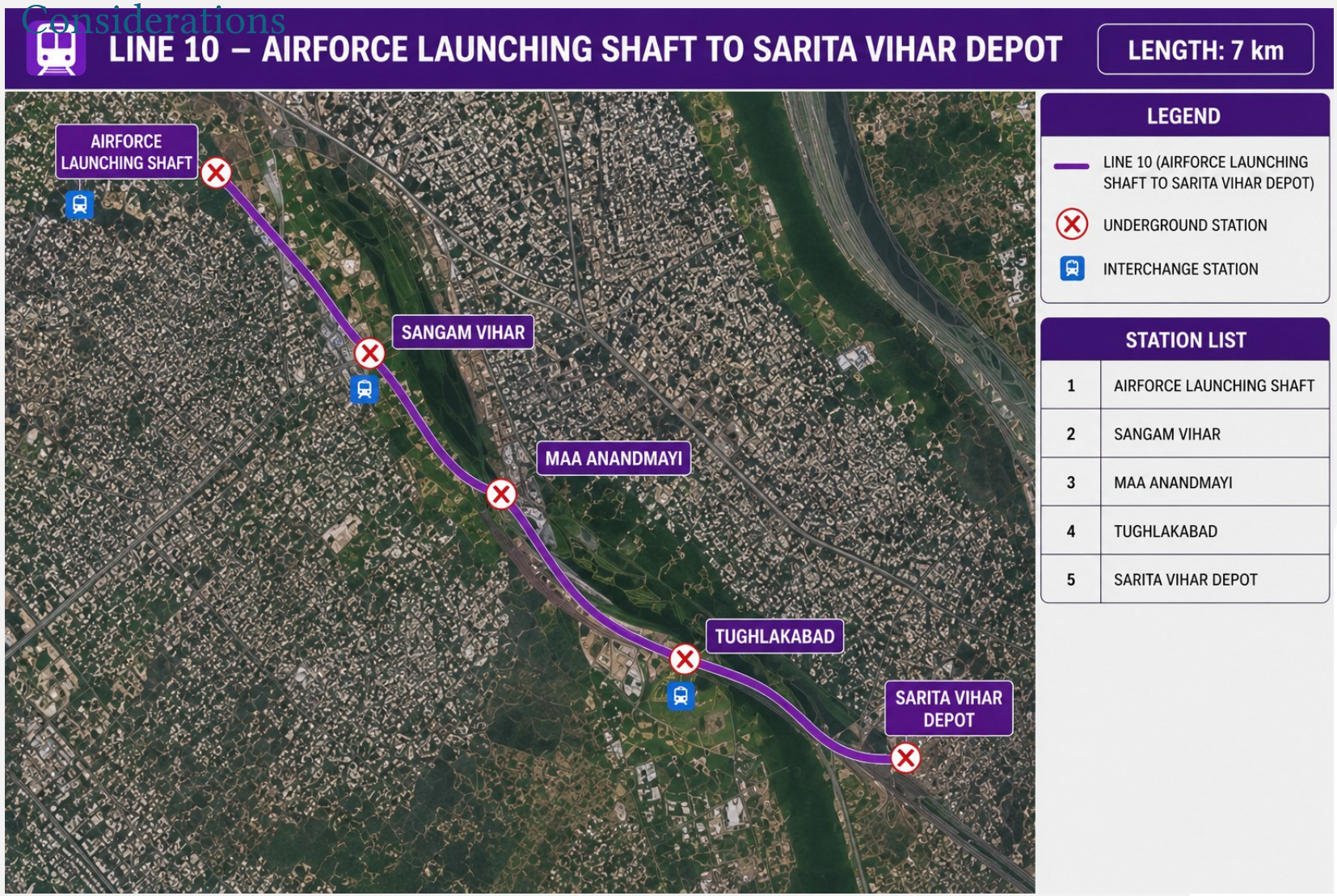


- Total Length: 12.5 Route Kms
- Alignment through urban residential and institutional zones
- Significant tunnel depth : ~25m to 38m near Chhattarpur Mandir to Neb Sarai
- Study required for Assessment of **Building and depth specific - vibration attenuation behaviour**



Under Ground Corridors

Line-10 (Airforce Launching Shaft to Sarita Vihar Depot): Key Engineering Considerations



- Total Length: ~7 Route Kms
- ~40% alignment through forest land
- Remaining through densely populated areas
- Required enhanced vibration assessment and targeted mitigation planning



Building Survey

Detailed Identification of Critical Receptors

- Physical survey are carried out **within 100m wide strip** along the alignment.
- Critical **Buildings are identified and categorized**, based on:
 - their functions (Category-1, Category-2 , Category-3 as per RDSO) and;
 - their constructions (Fragile/Historical Buildings, Low Story Buildings and Multiple Story Buildings as per Swiss Building Classifications).
- Nearest building in each category is selected for instrumentation and measurement.
- Impact of Vibration on such building is assessed and planning is done towards mitigation the excessive vibration.



Building Survey Contd.

Data Collection Format for Line -08 Extn.

S.No	Chainage	Building Name	Floors	Horizontal Distance From Track (m)	RDSO Category
1	23,300	Amazon Warehouse		0	3
2	23,350	Mahavir Sr. Model School		4	2
3	23,400	Hutments		0	2
4	23,450	Hutments		0	2
5	23,500	Residential building	G+4	20	2
6	23,550	Residential Building Mittal Bhavan	G+2	10	2
7	23,600	Residential Building Mittal Bhavan	G+4	0	2
8	23,650	Capital Nursing	G+2	16	1
9	23,650	Medicine Store	G+2	5	3
10	23,700	Residential	G+2	0	2
11	23,750	UC Building	G+4	0	2
12	23,800	Hutments		30	1
13	23,850	Residential	G+3	0	2
14	23,900	Ladies Jain Temple	G+2	0	1
15	23,950	Residential	G+3	0	2
16	24,000	Residential	G+3	0	2
17	24,050	Residential	G+5	0	2
18	24,100	Residential	G+3	0	2
19	24,150	Residential	G+2	0	2
20	24,200	Residential	G+4	0	2
21	24,250	Aman Battery Shop	G+4	0	2
22	24,300	Residential	G+2	0	2
23	24,350	Residential	G+2	0	2
24	24,400	Residential	G+2	0	2
25	24,450	Residential	G+2	0	2
26	24,500	Dental Clinic	G	30	1
27	24,550	ENT Clinic	G+2	30	1
28	24,600	Art Bhawan		0	1
29	24,650	Krishna Trading	G+3	0	3
30	24,700	Chaudhary Misthan Bhandar	G+2	0	3
31	24,750	Residential	G+2	0	2
32	24,800	Om Sweet Corner	G+2	5	3
33	24,850	Bala Ji Cycle	G+2	5	3
34	24,900	Residential	G+2	11	2
35	24,950	Shop	G+2	5	3

36	25,000	Commercial Building	G+3	5	3
37	25,050	Amba Cinema Hall		23	3
38	25,100	Oriental Insurance	G+2	5	3
39	25,150	Vijay Bank	G+3	15	3
40	25,200	HDFC Bank	G+2	10	3
41	25,250	Shop	G+2	13	3
42	25,300	Shop	G+3	5	3
43	25,350	Jain Clinic	G+2	20	1
44	25,400	Bedi Bakers	G+1	0	3
45	25,450	Residential	G+2	0	2
46	25,500	Temple		15	1
47	25,550	Forward Institute of Education	G+1	0	2
48	25,600	Forward Nursery School	G+2	0	1
49	25,650	Shop	G+2	14	3
50	25,700	Residential	G+3	0	2
51	25,750	Residential	G+3	0	2
52	25,800	Residential	G+2	0	2
53	25,850	Residential	G+2	0	2
54	25,900	Residential	G+2	0	2
55	25,950	Residential	G+2	0	2
56	26,000	Church		19	1
57	26,050	Residential	G+3	30	3
58	26,100	Residential	G+3	30	3
59	26,150	Residential	G+3	30	3
60	26,200	Residential	G+3	30	3
61	26,250	Palace	G+2	10	1
62	26,700	Residential	G+3	0	1
63	26,750	Residential	G+3	0	1
64	26,800	Residential	G+3	0	1
65	26,850	Residential	G+3	0	1
66	26,900	Residential	G+3	0	1
67	26,950	Dispensary	G+3	0	1
68	27,000	Residential	G+3	0	1
69	27,050	Residential	G+3	0	1
70	27,100	Residential	G+3	0	1
71	27,150	Residential	G+3	0	1
72	27,200	Residential	G+3	0	1
73	27,300	Temple		0	1
74	27,350	Shop	G+4	0	1
75	27,400	Shop	G+4	0	1
76	27,450	Shop	G+4	0	1
77	27,500	Shop	G+4	0	1
78	27,550	Shop	G+4	0	1
79	27,600	Shop	G+4	0	1
80	27,650	Shop	G+4	0	1

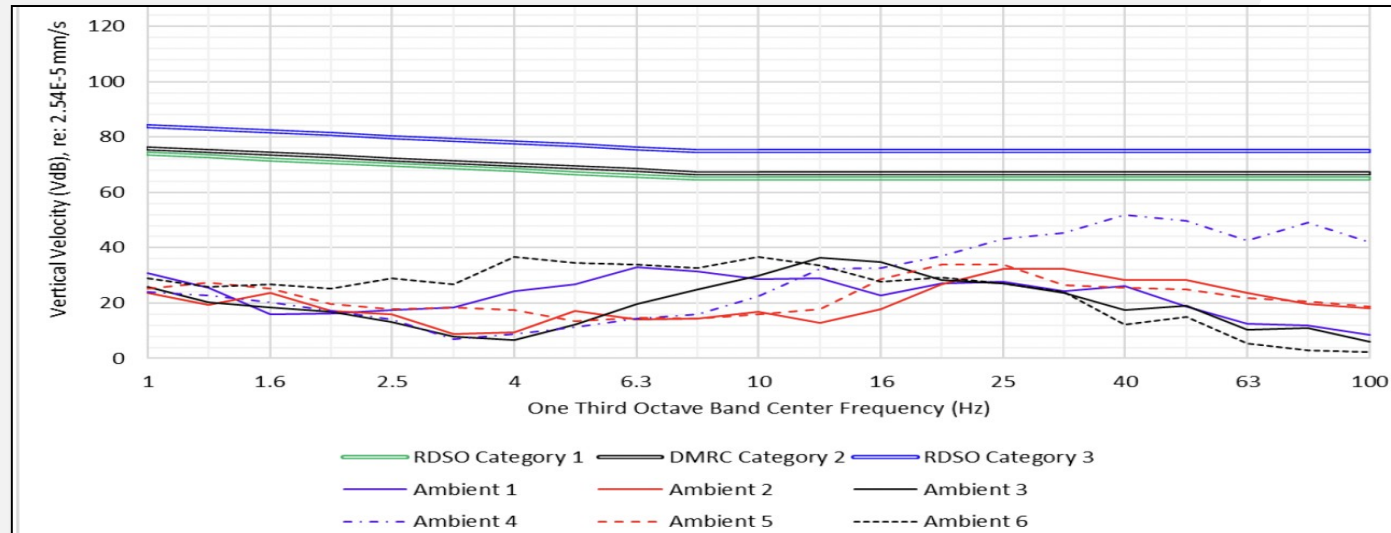
Ambient Vibration

Baseline Vibration Assessment

- Measurements conducted to capture **existing vibration environment**
- Identification of:
 1. High traffic zones
 2. Railway influence areas
- Measurements done at **06 locations in Line -08 Ext. and at 10 Locations in Line -10**
- ➔ Establishes **realistic baseline for comparison**



Results:





Source Vibration

Measurements of Train Borne Vibration at Source

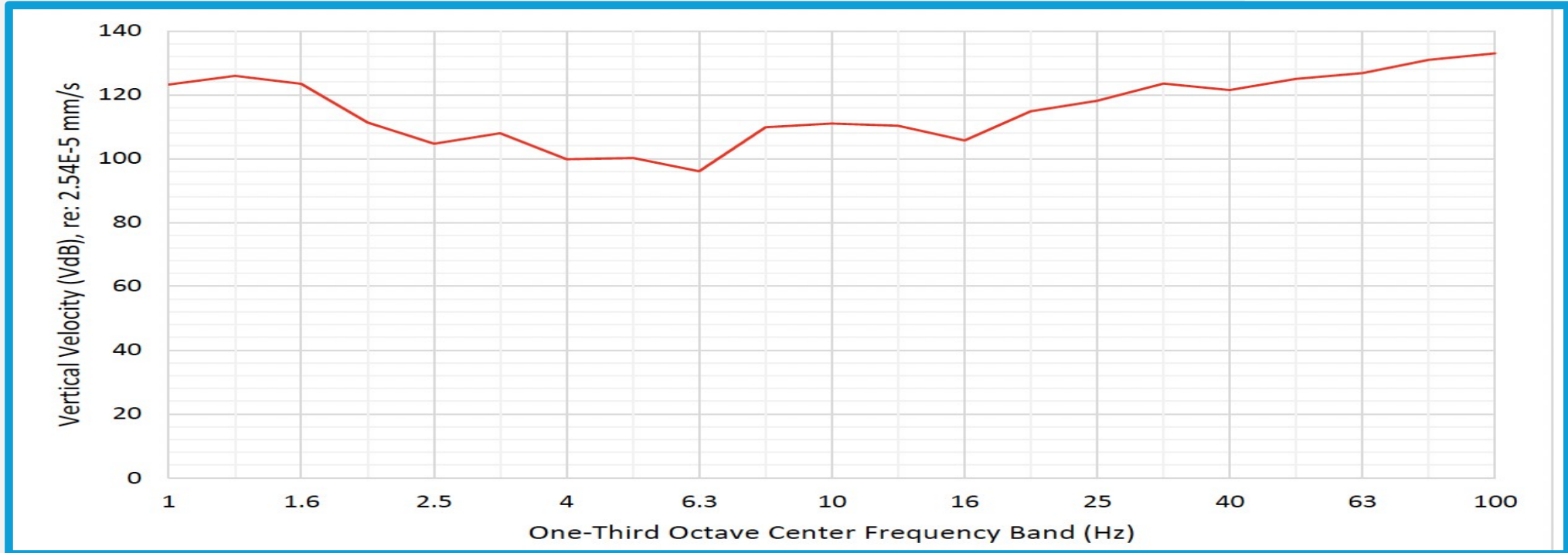
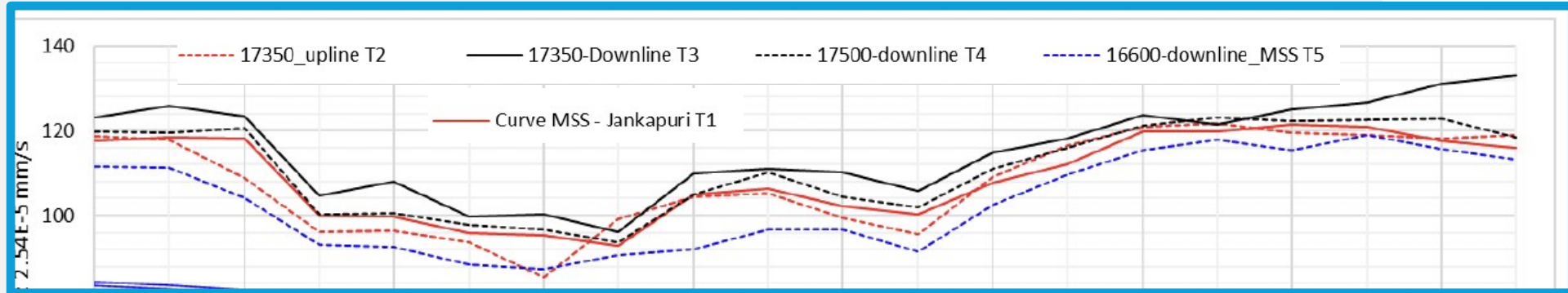
- Measurements carried out at **05 Locations** on **operational Line-08 of DMRC network**
- **Multiple scenarios** considered:
 1. Straight track / Curved Track (Speed)
 2. Station entry/exit (Acceleration/Breaking)
- **Equipment/Sensors** placed at:
 1. Rail (ii) Track bed (iii) **Tunnel wall**
- **Full Frequency Spectrum** was Measured instead of single Frequency.
 - ➔ **Maximum values at Tunnel wall adopted (conservative design)**





Source Vibration Contd.

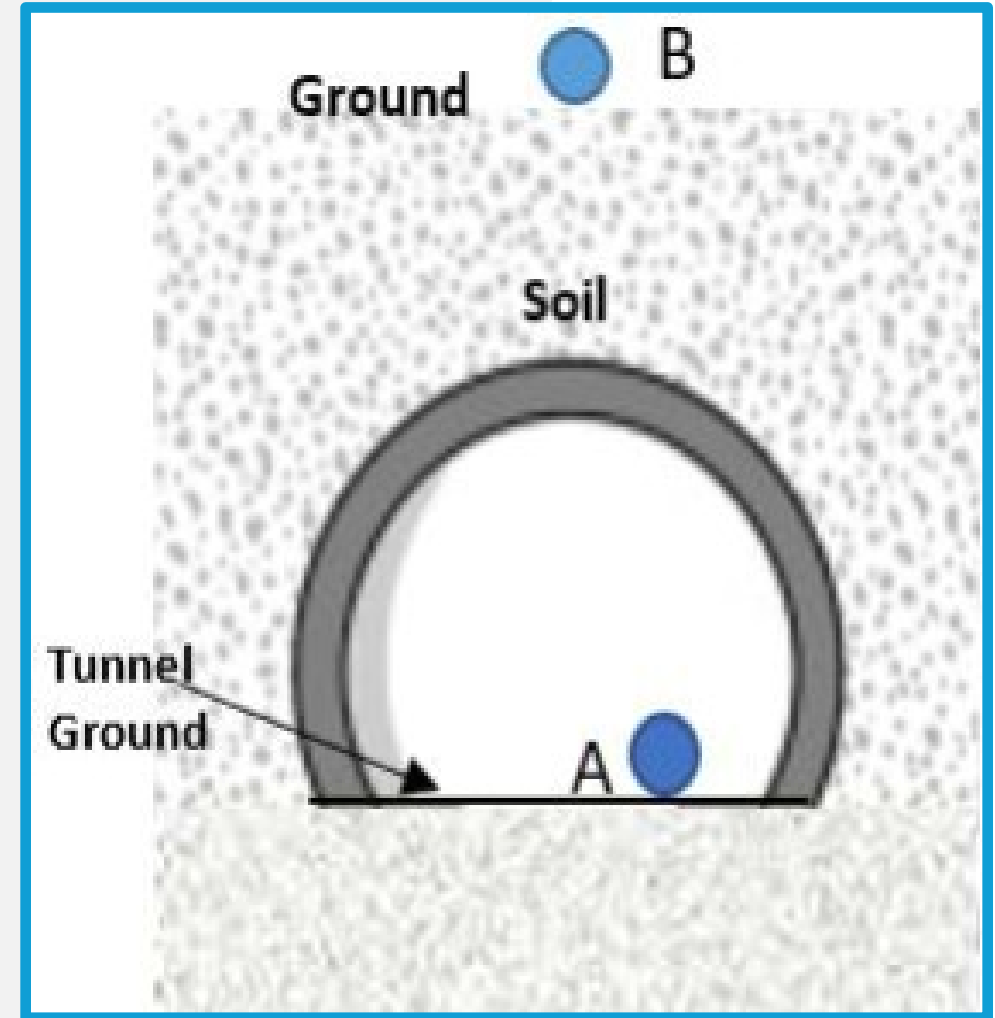
Measured Values – Tunnel Wall



Soil Transmission Loss

Tunnel-to-Soil Transmission Analysis

- Tunnel-to-Soil Attenuation is a function of:
 1. Distance from tunnel
 2. Soil type
 - Vibrations were generated at **A** and measured at **B** to assess Tunnel-to-Soil Loss.
 - **Measured at 04 Locations in Line-08 and 10 Locations in Line-10 inside the Constructed Tunnels.**
 - Locations selected based upon actual geotechnical data (Soil characteristics).
 - **Measured Loss correlated for other depths.**
- ➔ **Conservative values adopted for prediction**



Soil Transmission Loss Contd.

Measurement and Results



$$V = V_0 \cdot (D_0/D)^k$$

$$k = (\text{Log}V - \text{Log}V_0) / (\text{Log} D_0 - \text{Log}D)$$

Loss in VdB

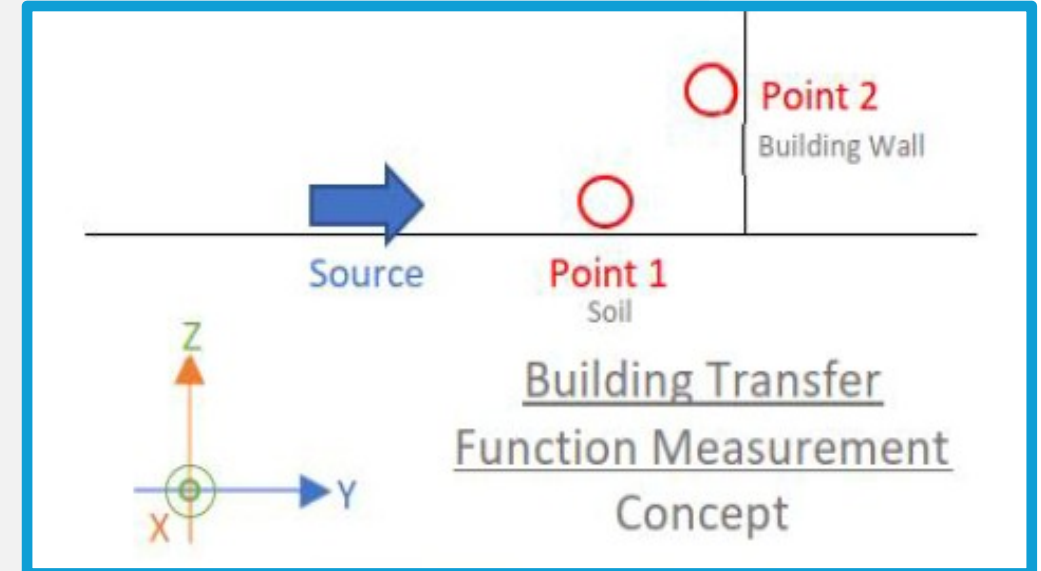
Radial distance from Crown of Tunnel



Building Transmission Loss

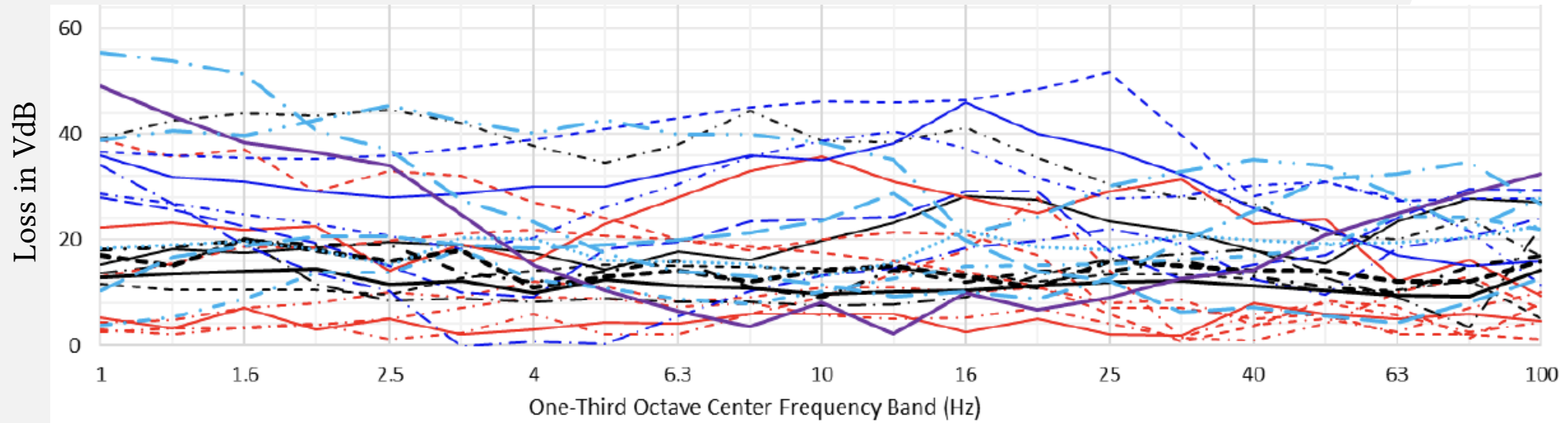
Building Response Assessment

- Transmission losses evaluated based on:
 1. Foundation type
 2. Structural configuration
- Vibrations were generated at Point 1 and measured at Point 2 to assess Soil-Building coupling Loss.
- Buildings were categorized. **Multiple measurements** conducted for each category.
- **26 Measurements at Line-08 and 35 Measurements at Line -10.**
- ➔ **Minimum attenuation values considered.**



Building Transmission Loss Contd.

Results



Frequency/ Vibration Loss	0.8	1.0	1.3	1.6	2.0	2.5	3.2	4.0	5.0	6.3	8.0	10.0	12.5	16.0	20.0	25.0	31.5	40.0	50.0	63.0	80.0	100.0
Category -3	-1	4	5	7	8	10	9	12	11	8	9	11	11	10	11	8	9	4	5	2	7	10
Category -2	2	3	3	3	4	5	7	9	9	7	8	6	5	5	7	4	1	6	8	6	2	4
Category -1	8	5	3	7	3	5	2	3	4	4	6	6	6	3	5	2	2	8	6	5	6	5



Selection of Vibration Mitigation System

Vibration Prediction Framework

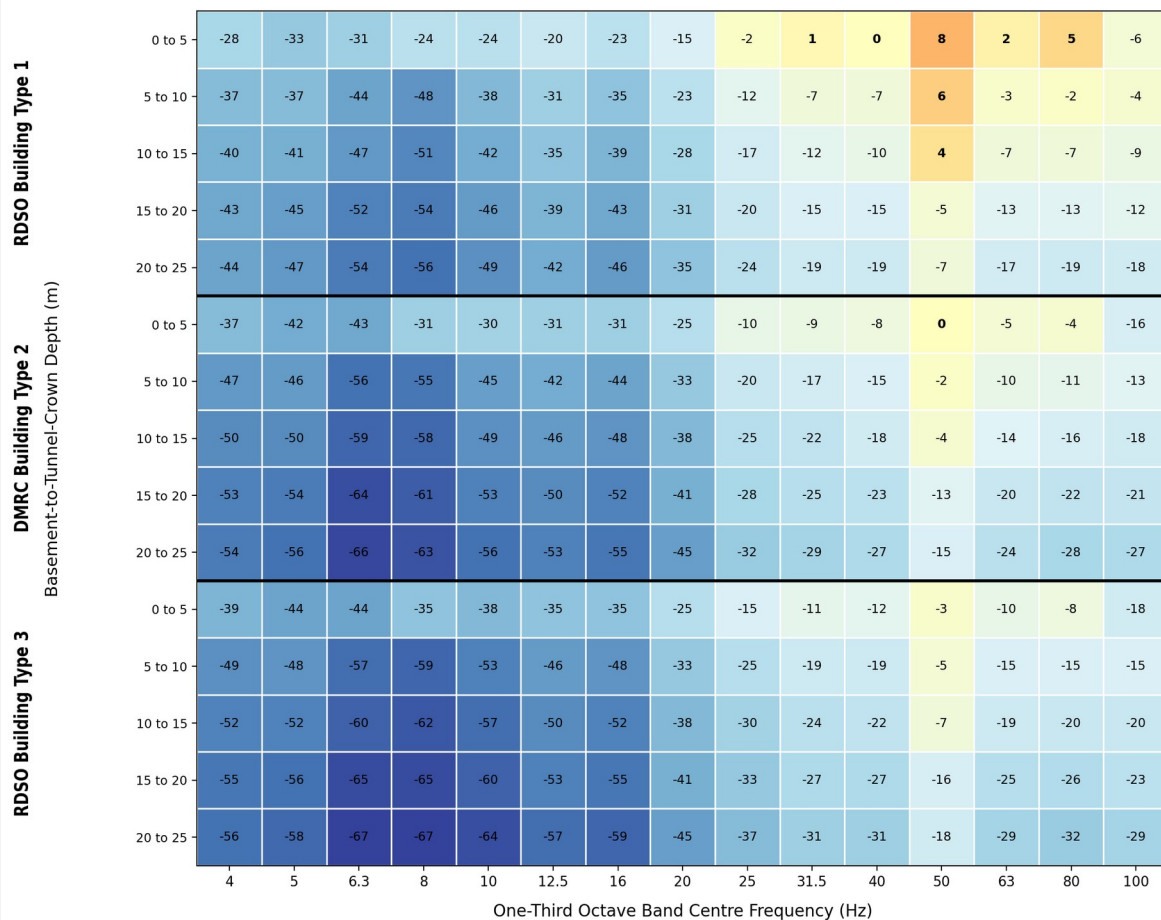
- Predicted Vibration at Building =
= Source Vibration (Max.) – Soil Transmission Loss (Min)
– Building Transmission Loss (Min.)
- **RDSO Vibration Criteria was reduced by 5 VdB i.e. 72 VdB to 67 VdB be kept as an additional margin.**
- Mitigation required = Predicted Vibration - Revised Criteria
- No Mitigation Required if Predicted Vibration is less than the Revised Criteria.



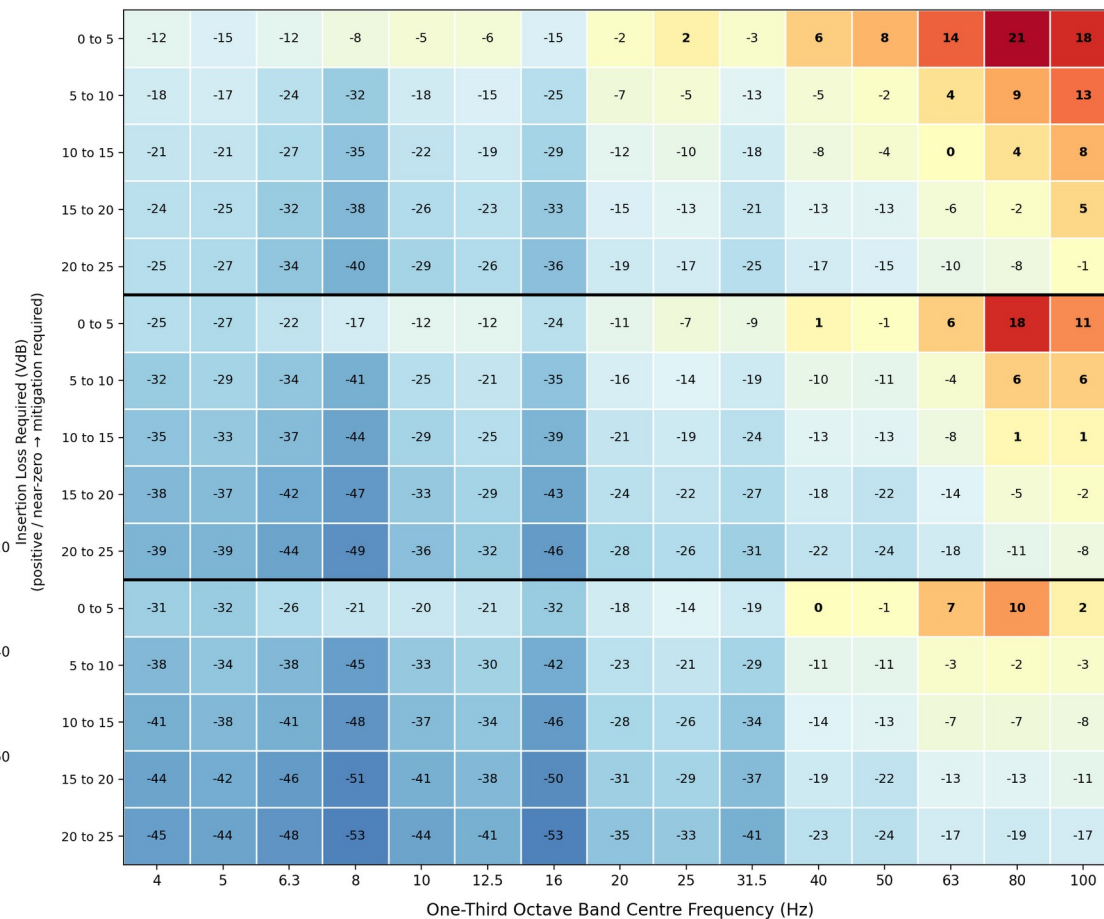
Selection of Vibration Mitigation System Contd.

Depth v/s Vibration Mitigation Requirement

**Table 7 — Insertion Loss Required for Isolation System
Derawal Nagar to Pulbangash (by depth, frequency band, and building type)**



**Table 8 — Insertion Loss Required for Isolation System
Pulbangash to R.K. Ashram (by depth, frequency band, and building type)**



Selection of Vibration Mitigation System Contd.

Location Specific Predicted Excessive Vibration

Ch.	Building Name	Floors	Depth (m)	Horizontal Distance from Track (m)	Final Distance (m)	RDSO Category	Isolation Required at 40 Hz (VdB)	Isolation Values (VdB)
27,100	Residential	G+3	-14.1	0	-14	1	0	8 (100 Hz)
27,150	Residential	G+3	-13.8	0	-14	1	0	8 (100 Hz)
27,200	Residential	G+3	-13.4	0	-13	1	0	8 (100 Hz)
27,300	Temple	-	-14.8	0	-15	1	0	8 (100 Hz)
27,350	Shop	G+1	-13.6	0	-14	1	0	8 (100 Hz)
27,400	Shop	G+1	-30.1	0	-30	1	0	0
27,450	Shop	G+3	-10.6	0	-11	1	0	8 (100 Hz)
27500	Sadar Bazar Metro Station							
27,500	Shop	G+4	-10.2	0	-10	1	0	13 (100 Hz)
27,550	Shop	G+4	-10.5	0	-11	1	0	8 (100 Hz)
27,600	Shop	G+5	-10.7	0	-11	1	0	8 (100 Hz)
27,650	Shop	G+3	-11.5	0	-12	1	0	8 (100 Hz)
27,700	Shop	G+4	-11.1	0	-11	1	0	8 (100 Hz)
27,750	Shop	G+4	-10.9	0	-11	1	0	8 (100 Hz)
27,800	Shop	G+4	-10.1	0	-10	1	0	13 (100 Hz)
27,850	Shop	G+4	-10.5	0	-11	1	0	8 (100 Hz)
27,900	Shop	G+4	-9.3	0	-9	1	0	13 (100 Hz)
27,950	Residential	G+3	-8.3	0	-8	1	0	13 (100 Hz)
28,000	Residential	G+3	-7.7	0	-8	1	0	13 (100 Hz)
28,050	Residential	G+3	-6.8	0	-7	1	0	13 (100 Hz)
28,100	Residential	G+3	-6.2	0	-6	1	0	13 (100 Hz)
28,150	Residential	G+3	-5.4	0	-5	1	6	21 (80 Hz)
28,200	Residential	G+3	-4.8	0	-5	1	6	21 (80 Hz)
28,250	Residential	G+3	-3.9	0	-4	1	6	21 (80 Hz)
28,300	Residential	G+3	-3.2	0	-3	1	6	21 (80 Hz)
28,350	Masjid	-	-2.6	30	-30	1	0	0
28,400	Shop	G+1	-2.2	0	-2	3	0	10 (80 Hz)

128	20650	Residential Building	G+1	0	18	13	2	13	-5	3(63 Hz)
129	20700	Residential Building	G+3	0	18	13	2	13	-5	3(63 Hz)
130	20750	Water Tank		0	19	14	3	14	-13	-5(63 Hz)
131	20800	Residential Building	G+4	0	20	15	2	15	-17	-1(63 Hz)
132	20860	Shiva Temple	G+0	12	19	14	3	18	-25	-9(63 Hz)
133	21100-21340	Railway Line & Highway		0	25	19		19	-35	-11(63 Hz)
Tughlakabad (21342-21600)										
134	21950-22100	Slum Area (dilapidated Condition)		0	20	15	2	15	-17	-1(63 Hz)
135	22150	Kendriya Vidyalaya NTPC	G+1	20	17	12	2	23	-27	-3(63 Hz)
136	22450	Residential Building (dilapidated Condition)	G+2	0	16	10	2	10	-5	3(63 Hz)
137	22500	Residential Building (dilapidated Condition)	G+3	0	15	10	2	10	-5	3(63 Hz)
138	22550	Residential Building (dilapidated Condition)	G+2	0	15	10	2	10	-5	3(63 Hz)
139	22600	Temple (dilapidated Condition)	G+0	0	15	10	2	10	-5	3(63 Hz)
140	22650	Residential Building (dilapidated Condition)	G+3	0	15	10	2	10	-5	3(63 Hz)
141	22700	Residential Building (dilapidated Condition)	G+3	0	15	10	2	10	-5	3(63 Hz)



Selection of Vibration Mitigation System Contd.

Recommended Measures

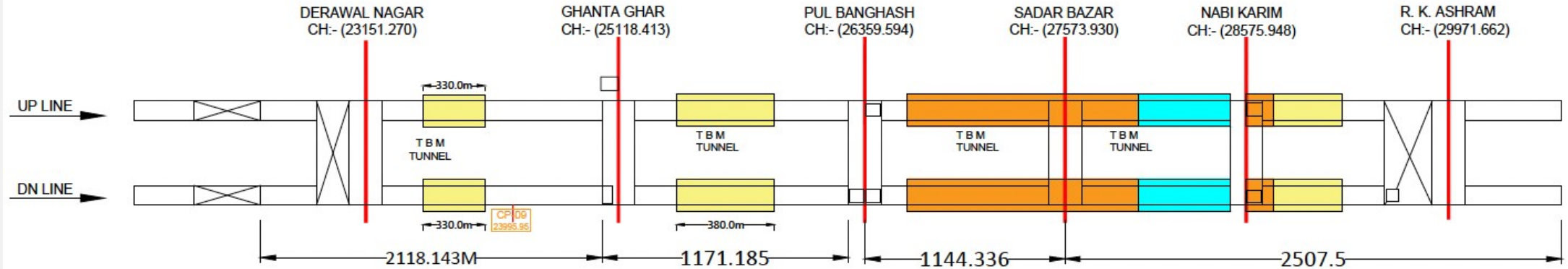
27,500	Sadar Bazaar					
27,500	Shop	-10	1	13(100 Hz)	NRFS1	RFST1/RFST2
27,550	Shop	-11	1	8 (100 Hz)		
27,600	Shop	-11	1	8 (100 Hz)		
27,650	Shop	-12	1	8 (100 Hz)		
27,700	Shop	-11	1	8 (100 Hz)		
27,750	Shop	-11	1	8 (100 Hz)		
27,800	Shop	-10	1	13(100 Hz)		
27,850	Shop	-11	1	8 (100 Hz)		
27,900	Shop	-9	1	13(100 Hz)		
27,950	Residential	-8	1	13(100 Hz)		
28,000	Residential	-8	1	13(100 Hz)		RFST3/RFST4
28,050	Residential	-7	1	13(100 Hz)		
28,100	Residential	-6	1	13(100 Hz)		
28,150	Residential	-5	1	21 (80 Hz)		
28,200	Residential	-5	1	21 (80 Hz)		
28,250	Residential	-4	1	21 (80 Hz)		
28,300	Residential	-3	1	21 (80 Hz)		
28,350	Masjid	-30	1	0		
28,400	Shop	-2	3	10 (80 Hz)		
28,450	Temple	-3	1	21 (80 Hz)		
28,500	Old Building	-8	1	13(100 Hz)		
28,500	Nabi Karim					

Isolator Reference	Manufacturer	Manufacturer Isolator Type
Full Surface MSS (RFST1)	Option 1	25mm Full surface MSS with natural frequency of 14-22 Hz
Strip + Filler MSS (RFST2)	Option 2	25mm Strip + Filler MSS with natural frequency of 12-22 Hz
Strip + Filler MSS (RFST3)	Option 3	37.5mm Strip + Filler MSS with natural frequency of 12-22 Hz
Discrete PUR pads with Filler Material (RFST4)	Option 4	300 mm square, discrete pads together with filler with natural frequency of 15-16Hz
NRFS 1	Option 5	336 Rail Fastening System
Delcor (NRFS2)	Option 6	Delcor Egg RF167



Recommended Mass Spring Systems

Line -08 Extn.

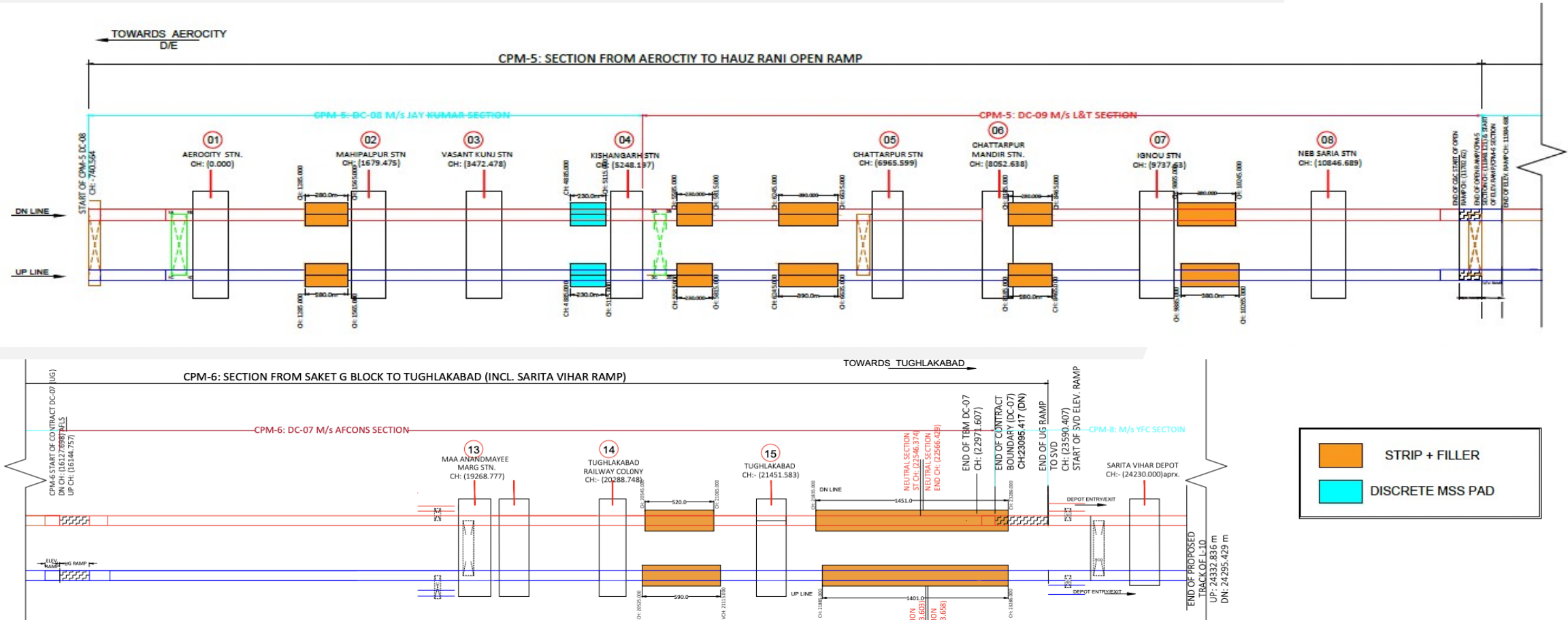


1. Total Underground length = 14.8 Track Kms
 2. MSS Planned:
 - i. Full Surface = 2.9 Track Kms
 - ii. Strip Plus Filler = 3.1 Track Kms
 - iii. Discrete Bearing = 1.3 Track Kms
- Total MSS Length = **7.3 Track Kms (approx. 50%)**



Recommended Mass Spring Systems Contd.

Line -10



1. Total Underground length = 39 Track Kms
2. MSS Planned: **Strip Plus Filler** = 7.4 Track Kms, **Discrete Bearing** = 0.5 Track Kms
3. Total MSS Length = **7.9 Track Kms (approx. 20%)**





Post-Installation Measures: Rail Grinding

- In addition to the mass-spring and resilient-fastening systems installed along the corridor, **rail grinding** is carried out post-installation to further control noise and vibration.
- DMRC operates **two dedicated rail grinding machines** one for broad gauge and one for standard gauge covering the full network.
- Grinding restores rail **surface quality** and removes rolling-contact fatigue and other surface **defects** that would otherwise roughen the wheel-rail interface.
- **Asymmetric rail profiles** are implemented during grinding to optimise the wheel-rail contact patch, **reducing noise and vibration** generated during train operation.





Rail Corrugation

SPEED + WAVELENGTH = FREQUENCY

This is the key relationship: a wheel moving at speed v over a corrugation of wavelength λ gets excited at frequency:

$$f = \frac{v}{\lambda}$$

Where:

v = Train speed (m/s)

λ = Corrugation wavelength (m)

f = Excitation frequency (Hz)

TYPICAL EXCITATION FREQUENCIES (METRO SPEEDS)

Train Speed (km/h)	20 mm Wavelength	40 mm Wavelength	60 mm Wavelength	80 mm Wavelength
60 km/h (16.7 m/s)	835 Hz	418 Hz	279 Hz	209 Hz
70 km/h (19.4 m/s)	970 Hz	485 Hz	323 Hz	242 Hz
80 km/h (22.2 m/s)	1,110 Hz	555 Hz	370 Hz	278 Hz



- Rail corrugation is a wave like pattern of peaks and troughs that develops on the rail running surface over time.
- Typical short pitch corrugations are most often associated with Noise and Vibrations

EXAMPLE (Typical Metro Speeds)

At ~70 km/h (19.4 m/s):

$\lambda = 20 \text{ mm}$

$f \approx 970 \text{ Hz}$

$\lambda = 80 \text{ mm}$

$f \approx 242 \text{ Hz}$

→ Train Direction →



Pre and Post Grinding

DLC_31_Line 1_MMOPL



Post Grind



DLC_27_Line 1_MMOPL



Post Grind



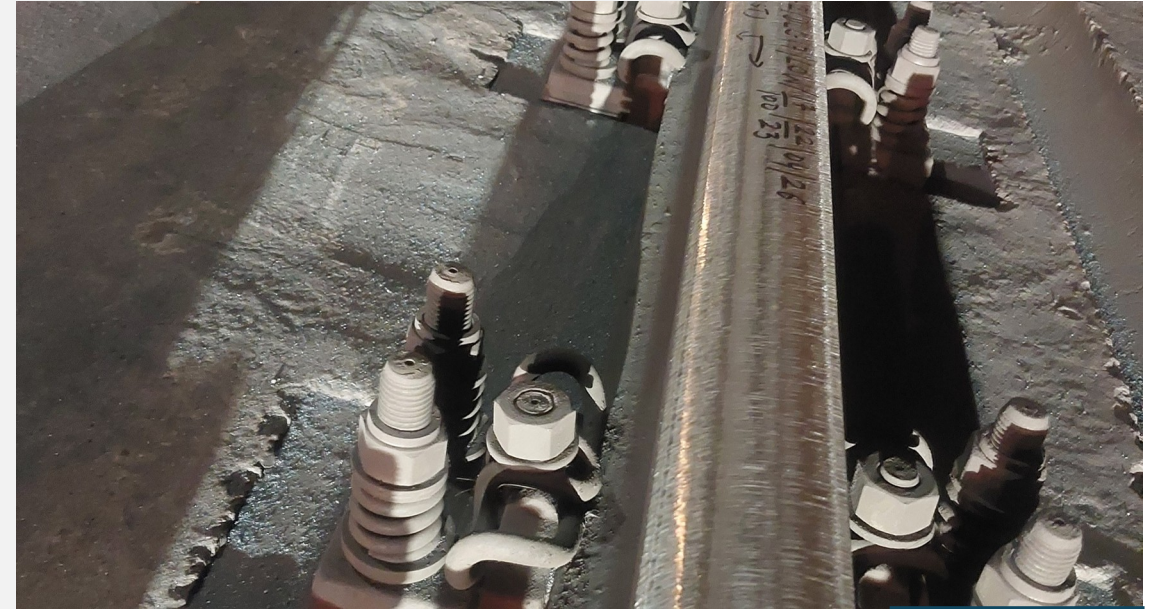


Pre and Post Grinding

DLC_17_Line 2_DMRC



Pre Grind



Post Grind

- **Post-grinding inspection confirmed successful removal of the short-pitch corrugation** previously present on the rail running surface, with the wave-like peak-to-trough pattern eliminated across the treated sections
- **The rail head transverse profile was successfully restored** to the design/target profile, correcting the wheel-rail contact geometry that had drifted due to corrugation-related wear
- **Asymmetric profile correction was successfully applied on curved sections**, re-establishing the intended high-rail/low-rail contact bands



Inferences

- **Vibration transmission from rail to receiver is predictable and repeatable** across the five-stage measurement framework (ambient → source → tunnel-to-soil → STF → BTF)
- **Mitigation is required across ~50% of Line 8** (7.3 of 14.8 track-km) **but only ~20% of Line 10** (7.9 of 39 track-km) → the lower requirement on Line 10 is attributable in part to the greater tunnel depth along that corridor, which increases the natural soil attenuation before vibration reaches the surface, reducing the insertion loss demanded of the isolation system
- **All reported values are real-time field measurements, not simulated outputs** → ambient, source, tunnel-to-soil, STF and BTF data were all obtained through direct instrumentation on operational and under-construction corridors, giving the prediction framework a measurement-based foundation rather than a purely analytical one
- **A future validation strategy is planned to verify whether the current implementation is working as intended** → post-installation field measurements on the completed sections will confirm whether the installed mass-spring systems and rail fastening are delivering the predicted insertion loss in practice, and will inform any refinement needed before the remaining sections are finalized
- **Rail grinding is being deployed as an additional, complementary mitigation measure** alongside the mass-spring and resilient-fastening systems → improving rail surface quality and introducing asymmetric profiles to further reduce noise and vibration at source



THANK YOU!